Substrate Carries, Architecture Pays: Check-In-Level Embeddings for Multi-Task POI Prediction

Anonymous Author(s)

Anonymous Institution
anonymous@example.org

Abstract. Multi-task learning for point-of-interest prediction typically pairs joint next-category and next-region heads over a shared graph embedding, assuming joint training transfers signal across tasks. We swap the per-place substrate for a check-in-level contextual graph embedding (one vector per visit) and run a controlled five-state Gowalla ablation under user-disjoint folds. The substrate is task-asymmetric: per-visit context lifts single-task next-category macro-F1 by 14 to 29 percentage points at every state (paired Wilcoxon p of 0.0312), while per-place embeddings tie or marginally exceed it under matched-head ceilings on next-region. Joint multi-task training over a cross-attention backbone then delivers a category lift at four of five states; the cost is a sign-consistent 7 to 17 percentage point drop on next-region top-10 accuracy, the textbook tradeoff. A per-visit counterfactual at Alabama attributes about 72 percent of the substrate gain to per-visit context.

Keywords: Multi-Task Learning · Point-of-Interest Prediction · Graph Embeddings.

1 Introduction

Location-Based Social Networks (LBSNs) record user check-ins at scale, motivating two recurring Point-of-Interest (POI) prediction tasks: the next visit's semantic category, which feeds content-aware recommendation and intent inference, and its spatial region, which feeds demand forecasting and short-horizon mobility planning. The two tasks place disjoint demands. **Category prediction** is a coarse semantic decision shaped by per-visit context (intent, time-of-day, co-location). **Region prediction** is a fine-grained spatial ranking that hinges on spatial-temporal regularities and short-horizon transitions. The multi-task learning (MTL) appeal is standard: a shared representation should let one head's signal transfer and lift both [2,17].

Two recent works frame the bottleneck. The first reported hard-parameter-sharing MTL on POI-stable graph embeddings (DGI [18]) yielding only marginal gains within standard deviations; the diagnosis was representation mismatch [15]. A follow-up decomposed the monolithic embedding into spatial, temporal, and

categorical sub-encoders, recovering category-side gains across three states [13]. Together these suggest embedding choice, not the MTL recipe, is the load-bearing factor.

Both prior responses treat the embedding as a *what* axis. We test an orthogonal *granularity* axis: emission rate. POI-stable embeddings like HGI [5] emit one vector per place; a check-in-level variant emits one per visit, capturing intent shifts across two visits to the same café at different hours. Our hypothesis is that **per-visit context** is the load-bearing substrate property for the easier head, and that once the substrate is fixed, joint training behaves like the textbook MTL tradeoff. The carry is task-asymmetric: per-visit *carries* next-category; per-place embeddings tie or marginally exceed it on next-region.

We adopt Check2HGI (a check-in-level extension of the hierarchical graph-infomax line emitting 64-dim per-visit vectors) and run a controlled ablation on five Gowalla [4] U.S.-state splits, Alabama (AL), Arizona (AZ), Florida (FL), California (CA) and Texas (TX), under user-disjoint folds. We compare Check2HGI against HGI under matched-head single-task (STL) baselines, and run a cross-attention multi-task backbone with per-task input streams, a GRU category head, and a STAN-based region head with an additive log-transition prior [9]. Matched-head STL ceilings let us separate substrate effects from head-capacity effects, a per-visit counterfactual decomposes the substrate gain itself, and an encoder-frozen isolation protocol surfaces a cross-attention pitfall in loss-side ablations.

The main contributions of this paper are as follows:

- **C1. A controlled five-state matched-head ablation that separates substrate from head capacity.** Check-in-level Check2HGI lifts single-task category macro-F1 over per-place HGI by +14.5 to +29 percentage points (pp) at every state (paired Wilcoxon $p = 0.0312$ each, 5/5 folds positive); on next-region the same substrate ties or marginally trails per-place embeddings. A pooled-vs-canonical counterfactual replicates at two states (Alabama and Arizona), attributing ∼72 % and ∼64 % of the category gap to per-visit context.
- **C2. A direct measurement of the multi-task tradeoff under the load-bearing substrate.** With the substrate fixed, joint cross-attention multi-task training adds a small category lift at four of five states and pays a sign-consistent 7 to 17 pp cost on next-region Acc@10 at every state. Drop-in adaptive balancers and a head-side fix do not close the gap, indicating the cost is structural to the cross-attention interaction rather than a loss-recipe artefact.

Sections 2, 3, and 4 cover related work, method, and setup. Section 5 reports substrate ablation, MTL-vs-STL, Δ_m, and external baselines. Section 6 presents the counterfactual, drop-in ablations, and the cross-attention note. Sections 7 and 8 close.

2 Related Work

Three lines of work bear on our study: graph-based POI embeddings that supply the substrate, multi-task learning balancers that sit on top of any joint architecture, and prior multi-task work on POI prediction tasks. We summarise each line briefly and position our contribution against it; a fourth note on cross-attention multi-task models is deferred to the methodological discussion in Section 6.

2.1 Graph-based POI embeddings

Substrate choice is the load-bearing axis our study tests. Early POI representations learn one vector per place from co-visit statistics, treating the POI as the atomic unit, and category-aware extensions inject category supervision into the same objective. Self-supervised graph contrastive methods then sharpened the representation: Deep Graph Infomax (DGI) [18] maximises mutual information between local node embeddings and a global graph summary, and Hierarchical Graph Infomax (HGI) [5] adapts the contrast across POI, region, and city graphs to capture multi-scale urban structure. DGI and HGI both emit *POI-stable* vectors: the same place visited twice yields the same representation. The substrate we use, *Check2HGI*, is a check-in-level extension of HGI's contrastive objective: contrastive views are sampled at the granularity of a single check-in, and the visit's temporal and co-visit context perturbs the view, so the same POI visited twice can yield two distinct vectors. This per-visit emission rate is the property our substrate axis tests; the recipe is described in supplementary material.

2.2 Multi-task learning balancers

A second body of work studies how to combine task losses or gradients once the architecture is fixed. Gradient-balancing methods rescale, project, or align per-task gradients (e.g., NashMTL [12] solves a per-step bargaining game); more recent proposals continue along the same axis, with FAMO [8] adapting loss weights from per-task progress and Aligned-MTL [14] aligning gradient directions through an independent-component decomposition. The dense-prediction survey [17] aggregates the line and reports that no balancer dominates across task pairs; the textbook tradeoff identified by Caruana [2] and quantified by Maninis et al. [10] through the Δ_m joint score remains the operating regime. Our setup uses a fixed static-weight aggregation with $\lambda_{\text{cat}} = 0.75$. We do not propose a new balancer, and we ablate against FAMO and Aligned-MTL in Section 6 to check whether a drop-in balancer recovers the next-region cost we will report.

2.3 Prior POI multi-task work

A third line is concept-aligned with our task pair. HMT-GRN [7] jointly predicts next-POI and next-region through a hierarchical graph recurrent network,

GETNext [20] combines a transformer with an additive log-transition prior over a trajectory flow map, and the human-mobility representation model [3] jointly models attribute and trajectory signals through shared encoders. STAN [9] addresses the next-location subproblem with spatio-temporal self-attention and is the backbone we adapt for our region head; we borrow GETNext's additive $\alpha \cdot \log T$ prior pattern but do not include its friendship or check-in graph priors, so our region head is not a faithful GETNext reproduction. ReHDM [6] extends the hypergraph line with a region-aware dual-level model and serves as our concept-aligned external next-region baseline. Two recent works on the precise category and next-POI pairing are closer still. A first study trained a hard-parameter-sharing multi-task framework over POI-stable graph embeddings (DGI), comparing against single-task baselines such as HMRM [3] and MHA+PE [21], and reported only marginal joint gains, with most differences within standard deviations; the diagnosis pointed at representation mismatch, where a single shared encoder learns a compromise specialised enough for neither head [15]. A follow-up tested an embedding-side response by decomposing the monolithic graph embedding into independent spatial, temporal, and categorical sub-encoders, and recovered substantial category-side gains across three states, supporting the reading that the embedding choice was the load-bearing factor [13]. Both responses kept the embedding's per-POI emission rate fixed and varied the *what* axis (which features to encode); our study varies the *granularity* axis instead.

Our work re-examines a default reading of multi-task learning for POI prediction: that joint training transfers signal between heads to lift both. With the substrate held fixed at the right granularity, joint cross-attention training delivers a small additional category lift at four of five Gowalla [4] state splits and pays a sign-consistent cost on next-region at every state. We use a controlled five-state ablation on user-disjoint folds to reattribute the joint result, separating the substrate's contribution from the architecture's.

3 Method

We describe the substrate (adopted), the backbone (the object of study), and the training protocol. Notation: a check-in is $c = (p, r, t)$ with POI p, region r, timestamp t; trajectories split into non-overlapping length-nine windows predicting the category and region of the tenth check-in ($L = 9$).

3.1 The Check2HGI substrate

Check2HGI extends the hierarchical graph-infomax line of HGI [5] (built on DGI [18]) by inserting a check-in level beneath the POI level: the hierarchy becomes Check-in $\rightarrow$ POI $\rightarrow$ Region $\rightarrow$ City. Bottom-level nodes are individual visits; edges follow each user's chronological visit sequence with a temporal-decay weighting, and node features carry the visit's category and timestamp. Embeddings propagate upward by attention pooling (check-ins to POIs), a region-graph convolution, and area-weighted aggregation (regions to city); mutual information

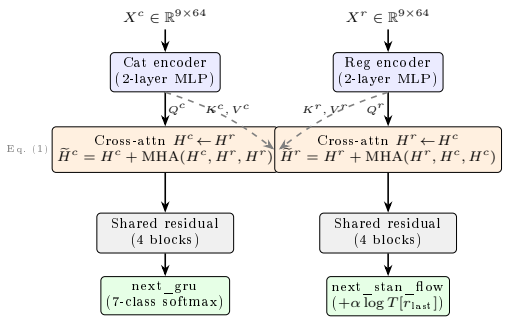

Fig. 1. Cross-attention multi-task backbone. Two task-specific encoders project per-task input streams X^c, X^r into H^c, H^r. A symmetric cross-attention block (Eq. 1) lets each stream query the other through key/value projections of the surviving stream; a four-block shared residual backbone then splits into the GRU category head and the STAN-based region head with additive log-transition prior. Dashed arrows mark the K/V cross-feeds responsible for the encoder co-adaptation discussed in Section 6.3.

is maximised across the three adjacent boundaries via bilinear discriminators against random-shuffle negatives. Two products feed our backbone after the encoder is frozen, both 64-dimensional: a per-check-in vector $\mathbf{e}_c$ from the bottom level for the category stream, and a per-region vector $\mathbf{e}_r$ from the third level for the region stream. We treat Check2HGI as a fixed input substrate; the full recipe is in supplementary material.

3.2 Cross-attention multi-task backbone

The backbone keeps two parallel input streams (one per task) and bridges them with cross-attention before a shared backbone splits into per-task heads. Let $X^c, X^r \in \mathbb{R}^{L\times d}$ feed the category and region sides at $L = 9$, $d = 64$. Two task-specific encoders (two-layer MLPs with LayerNorm and dropout) project each stream to $\mathbb{R}^{L\times d_m}$ at $d_m = 256$, yielding H^c and H^r.

A cross-attention block bridges the streams. With $\mathrm{MHA}(Q, K, V)$ scaled dot-product attention ($h = 8$),

$$\widetilde{H}^c = H^c + \mathrm{MHA}(H^c,\, H^r,\, H^r), \qquad \widetilde{H}^r = H^r + \mathrm{MHA}(H^r,\, H^c,\, H^c), \tag{1}$$

so each stream queries the other. The cross-attended streams then pass through a shared backbone of four residual blocks (LayerNorm, linear, LeakyReLU, dropout) before splitting into per-task heads (Figure 1). Eq. (1) is the architectural object the result tables interrogate.

The category head is a two-layer GRU over the cross-attended check-in stream with attention pooling and a seven-class softmax. The region head is an additive-prior STAN [9] variant: a STAN backbone over the cross-attended region stream, scored against learnable region embeddings, with an additive log-transition prior $\alpha \cdot \log T[r_{\mathrm{last}}]$ (learnable α, built per fold from training-fold edges

only). The log-prior pattern follows GETNext, but our head omits friendship and check-in graph priors and is not a faithful reproduction. The single-task ceilings reuse these head classes, so multi-task and single-task rows differ only in the shared backbone above them.

3.3 Training protocol

We aggregate the two task losses with a static weighting,

$$\mathcal{L} = \lambda_{\text{cat}}\,\mathcal{L}_{\text{cat}} + \lambda_{\text{reg}}\,\mathcal{L}_{\text{reg}}, \qquad \lambda_{\text{cat}} = 0.75, \quad \lambda_{\text{reg}} = 0.25, \tag{2}$$

where $\mathcal{L}_{\text{cat}}, \mathcal{L}_{\text{reg}}$ are unweighted cross-entropy, fixed across all MTL rows. FAMO [8] and Aligned-MTL [14] drop-in comparisons are in Section 6.

The optimiser is AdamW with three parameter groups (category, region, shared) and OneCycleLR, batch size 2048 (1024 at the largest state), gradient accumulation of two. Training runs 50 epochs in FP32; per-task best checkpoints select on validation Acc@10 (region) and macro-F1 (category) for epochs ≥ 5, since the region head saturates before the category head converges. Both heads use unweighted cross-entropy, matching the single-task ceilings in Section 4.

Folds, seeds, and multi-seed pooling follow the user-disjoint StratifiedGroupKFold setup of Section 4; the same splits are reused across substrates, heads, and recipes so paired tests are well defined. **A methodological caveat:** under cross-attention, a loss-side ablation that zeros one task's weight does not isolate the targeted encoder, since it co-adapts through the attention key and value projections of the surviving stream. We use encoder-frozen isolation (random-init, frozen) as the clean architectural decomposition throughout; Section 6.3 develops the note. Code, fold splits, seeds, per-fold transition priors, and regression tests are released at `https://anonymous.4open.science/r/PoiMtlNet-FE6A/`.

4 Experimental Setup

4.1 Datasets and preprocessing

We evaluate on five U.S.-state splits of the public *Gowalla* check-in dataset [4]: AL, AZ, FL, CA, TX. Each split is filtered to users with at least five check-ins, applied per state before fold creation so each state preserves its native cardinality. Per-state statistics are in Table 1; FL/CA/TX serve as the headline scale and AL/AZ as smaller-scale anchors. Region labels come from the per-state region partition used by the substrate engine.

For each user we form non-overlapping length-9 windows over the chronological sequence and predict the category and region of the tenth check-in (shorter sequences padded). Folds are **user-disjoint** via `StratifiedGroupKFold(groups=userid, shuffle=True, random_state=42)` with five folds, so train/val/test never share a user. Splits are reused across substrates, heads, and recipes so paired tests are well defined. The next-region transition prior is rebuilt per fold from training-fold edges only and seed-tagged, so the leak-free protocol holds at every comparison.

Table 1. Dataset statistics for the five Gowalla [4] U.S.-state splits, after filtering to users with at least five check-ins. The headline three (FL/CA/TX) sit above the rule; the smaller-scale anchors (AL/AZ) sit below. Region cardinality is the prediction target on the next-region task.

State	Users	Check-ins	POIs	Regions	Mean traj. length
Florida (FL)	13,935	1,392,262	76,266	4,703	99.9
California (CA)	26,171	3,148,594	168,771	8,501	120.3
Texas (TX)	27,603	4,067,543	160,575	6,553	147.4
Alabama (AL)	1,622	109,695	11,666	1,109	67.6
Arizona (AZ)	3,331	227,956	20,469	1,547	68.4

4.2 Baselines and metrics

The headline contrast is the cross-attention multi-task model against matched-head single-task (STL) ceilings: a GRU category head and a STAN-based region head with an additive log-transition prior over training-fold-only edges [9]. **The same head classes appear on both sides of the contrast, so any difference between the multi-task and single-task rows is attributable to the sharing architecture, not to head capacity.** External baselines are POI-RGNN and MHA+PE (category) and Markov-1-region, single-task STAN, and ReHDM (region); we report them in §5.

Category metrics: macro-F1 (primary) and Acc@1/Acc@3 (top-k accuracy, secondary, since the 7-way distribution is skewed). Region metrics: Acc@10 (primary), Acc@5 and MRR (mean reciprocal rank, secondary). The joint score is Δ_m [10,17] against the matched-head STL ceiling, reported on a primary pair (macro-F1, MRR) and a secondary pair (macro-F1, Acc@10).

4.3 Statistical protocol

Paired contrasts use the **paired Wilcoxon signed-rank** test [19] with `alternative='greater'` for directional claims and the two-one-sided test (**TOST**) for non-inferiority at $\delta \in \{2, 3\}$ pp. At the single-seed five-fold scale the max significance for $n = 5$ is $p = 0.0312$ (all folds agreeing).

Two ceiling protocols sit side by side. The **substrate axis** (Table 2) uses the seed-42 five-fold paired Wilcoxon at the $n = 5$ ceiling; multi-seed mean and seed-σ are shown for Check2HGI at AL/AZ/FL where multi-seed runs exist, but the paired test stays single-seed. The **architectural axis** (Tables 3, 4) pools across four extra seeds $\{0, 1, 7, 100\}$ on top of seed-42, yielding $n = 20$ multi-seed fold-pairs per state; the FL Δ_m pair additionally carries $n = 25$ fold-pairs from a second sweep. Effect sizes are in pp for F1, Acc@k, and MRR; Δ_m is in percent. Code, fold splits, seeds, and per-fold priors are in the anonymous snapshot referenced in §3.

5 Results

We report the empirical reattribution across five Gowalla state splits. Section 5.1 isolates the substrate axis (Check2HGI vs HGI under matched-head STL ceilings). Section 5.2 compares the cross-attention MTL backbone against those ceilings, the headline contrast. Section 5.3 reports the joint Δ_m score. Section 5.4 situates the model against external baselines.

5.1 Substrate ablation: per-visit context is task-asymmetric

Table 2(a) reports next-category macro-F1 under the matched-head `next_gru` STL ceiling, with substrate as the only changing factor. Check2HGI exceeds HGI by +14.52 to +29.02 percentage points (pp) at every state (smallest at AZ, largest at FL). The paired Wilcoxon signed-rank test reaches the $n = 5$ ceiling everywhere ($p = 0.0312$, 5/5 folds positive). Two robustness probes back this: the AL/AZ gap is head-invariant across linear, single-layer, GRU, and LSTM probes, and the FL/CA/TX gap replicates under the matched-head ceiling. We read the lift as evidence that per-visit context, not POI-level attribute encoding, is what the next-category head exploits.

Table 2(b) reports Acc@10 under the matched-head `next_stan_flow` STL ceiling. POI-stable HGI is nominally above Check2HGI by 1.6 to 3.1 pp at every state; on FL/CA/TX the gap shrinks to 1.6 to 2.1 pp and TOST non-inferiority at $\delta = 2$ pp passes at CA and TX. At FL the $\delta = 3$ pp test passes. AL and AZ stay outside both margins. The short-horizon transition structure driving the region head is captured equally well, or fractionally better, by per-place embeddings whose category-level pooling smooths the per-visit variance that helps the category head.

The substrate's effect is task-asymmetric. The category head reads per-visit context and lifts by an order of magnitude more than its standard-deviation envelope; the region head reads transition structure preserved under per-place pooling, tying within ±2 pp. Section 6 attributes the asymmetry through a per-visit-context counterfactual at Alabama. Any MTL system on this task pair operates on a substrate that already encodes the easier task more clearly than the harder one.

5.2 Multi-task vs single-task on the load-bearing substrate

Table 3 reports the headline contrast: the cross-attention MTL backbone against the matched-head `next_gru` and `next_stan_flow` STL ceilings, both on Check2HGI, multi-seed pooled at $n = 20$ fold-pairs per state. Category is positive at four of five states: FL +1.40 pp ($p = 2\text{e}{-6}$), CA +1.68 pp ($p = 2\text{e}{-6}$), TX +1.89 pp ($p = 2\text{e}{-6}$), AZ +1.20 pp ($p < 1\text{e}{-4}$). At AL the contrast is small-significantly negative ($\Delta = -0.78$ pp, $p = 0.036$, $n = 20$), under 2 % relative on the 41 % F1 scale. The four positive states sit outside the seed and fold envelope; we report the AL negative as is rather than absorbing it.

Table 2. Substrate ablation under matched-head single-task ceilings, five-fold cross-validation, seed-42 paired Wilcoxon at the $n = 5$ ceiling ($p = 0.0312$ when 5/5 folds agree). For Check2HGI at AL/AZ/FL the cell shows the mean over four extra seeds $\{0, 1, 7, 100\}$ with seed-σ for context; the paired test uses the seed-42 fold-pairs. (a) Next-category macro-F1 with a GRU category head; the asterisk marks 5/5 folds positive. (b) Next-region Acc@10 with a STAN-based region head and additive log-transition prior built per fold from training-fold edges; negative Δ values indicate POI-stable HGI is nominally ahead, and TOST verdicts at $\delta \in \{2, 3\}$ pp report whether the gap stays inside the non-inferiority margin (✓) or not (×). The architectural axis (Tables 3 and 4) reports a different ceiling estimator: see §4.3.

(a) Next-category macro-F1.

State	HGI cat F1	Check2HGI cat F1	Δ (pp)	Wilcoxon p
Alabama (AL)	25.26 ± 1.18	41.35 ± 0.17	+15.50*	0.0312
Arizona (AZ)	28.69 ± 0.79	43.90 ± 0.17	+14.52*	0.0312
Florida (FL)	34.41 ± 1.05	63.43 ± 0.98	+29.02*	0.0312
California (CA)	31.13 ± 1.04	59.94 ± 0.59	+28.81*	0.0312
Texas (TX)	31.89 ± 0.55	60.24 ± 1.84	+28.34*	0.0312

(b) Next-region Acc@10.

State	HGI Acc@10	Check2HGI Acc@10	Δ (pp)	TOST $\delta = 2$	TOST $\delta = 3$
Alabama (AL)	61.86 ± 3.29	59.15 ± 3.48	−2.71	×	×
Arizona (AZ)	53.37 ± 2.55	50.24 ± 2.51	−3.13	×	×
Florida (FL)	71.34 ± 0.64	69.22 ± 0.52	−2.12	×	✓
California (CA)	57.77 ± 1.12	55.92 ± 1.20	−1.85	✓	✓
Texas (TX)	60.47 ± 1.26	58.89 ± 1.28	−1.59	✓	✓

Next-region is the textbook tradeoff. MTL pays −7.34 pp (FL) to −16.59 pp (TX) on Acc@10 at every state, all paper-grade significant ($p \leq 1.9\text{e}{-6}$). The pattern is broadly downward on AL→AZ→FL, with CA preserving the FL regime and TX breaking the monotonic shrink. We read the TX non-monotonicity descriptively: state-specific factors beyond raw class count (geographic POI distribution, user-density skew, per-state log-transition prior coverage) plausibly contribute; we do not read the AL→FL contraction as an inferential scaling claim.

At FL the MTL row sits below the Markov-1-region floor on Acc@10 (63.27 vs 65.05, Table 4). The honest reading is that the short-horizon transition structure at FL's region cardinality is largely captured by a one-step Markov chain, and Acc@10 saturates near that floor for any model whose region head pays a substrate cost. The matched-head `next_stan_flow` STL ceiling on Check2HGI sits well above the floor at 70.62, so the cross-attention MTL backbone is trading that ceiling for the joint regime rather than failing to clear a Markov baseline outright. The Δ_m MRR axis at FL (+2.33 %, $n = 25$, paper-grade in Section 5.3) characterises the joint tradeoff more cleanly than Acc@10 in this saturation regime; we report both axes and read them jointly.

Table 3. Headline contrast on the Check2HGI substrate: cross-attention multi-task model vs. matched-head single-task ceilings. Multi-seed pooled at $n = 20$ fold-pairs per state (seeds $\{0, 1, 7, 100\}$, five user-disjoint folds). Cells are mean $\pm$ pooled σ; Δ is multi-task minus single-task in percentage points; paired one-sided Wilcoxon p-values (alternative "greater" for positive Δ, "less" for negative). State codes: FL/CA/TX (headline scale, above the rule), AL/AZ (smaller-scale anchors, below).

	Next-category (macro-F1)				Next-region (Acc@10)			
State	STL	MTL	Δ	p	STL	MTL	Δ	p
FL	67.16±0.13	68.56±0.79	+1.40	2e−6	70.62±0.09	63.27±0.10	−7.34	1.9e−6
CA	62.39±0.13	64.07±0.14	+1.68	2e−6	56.85±0.09	47.35±0.11	−9.50	1.9e−6
TX	63.11±0.13	65.00±0.11	+1.89	2e−6	59.44±0.09	42.84±0.14	−16.59	1.9e−6
AL	41.35±0.17	40.57±0.24	−0.78	0.036	61.21±0.18	50.17±0.24	−11.04	1.9e−6
AZ	43.90±0.17	45.10±0.19	+1.20	<1e−4	53.06±0.15	40.78±0.07	−12.27	1.9e−6

Δ in percentage points (multi-task − single-task).

With the substrate fixed at the per-visit granularity, the MTL backbone delivers a small category lift at four of five states and pays a sign-consistent cost on the harder head at every state, the classic MTL tradeoff. The cost is structural to the cross-attention shared-backbone interaction with the region head; Section 6 reports drop-in replacements (FAMO, Aligned-MTL, hierarchical-additive-softmax region head) that fail to close the next-region gap.

5.3 Joint Δ_m score: a Pareto-positive cell at Florida

We compute Δ_m [10,17] against the matched-head single-task ceiling. On the primary pair (category macro-F1, region MRR), FL carries an $n = 25$ multi-seed sweep and reaches Δ_m-MRR $= +2.33\,\%$ at $p = 2.98\text{e}{-8}$, 25/25 fold-pairs positive: a Pareto-positive cell on the primary axis. The secondary pair at FL is Δ_m-Acc@10 $= -1.12\,\%$ at $p = 3.20\text{e}{-5}$ (4/25); MRR rewards near-correct candidates that raw top-10 accuracy misses. The other four states report at the $n = 5$ Wilcoxon ceiling: Δ_m is negative on both axes (MRR from $-24.84\,\%$ to $-1.61\,\%$; Acc@10 from $-22.41\,\%$ to $-6.85\,\%$), at most 1/5 fold-pairs positive. FL is the only Pareto-positive cell on the primary axis at multi-seed.

5.4 External baselines

Table 4 situates the model against external baselines. On next-category, our matched-head single-task ceiling on Check2HGI sits well above our faithful POI-RGNN reproduction [1], but the gap should be read with care: the published POI-RGNN evaluation used non-user-disjoint folds, so absolute numbers are not directly comparable to our user-disjoint reproduction. The substrate-axis ablation of Section 5.1 attributes a substantial share of the lift to per-visit context, not to multi-task aggregation. On next-region, the faithful STAN reproduction [9] on HGI is the strongest external baseline; per-place HGI is a more natural fit for the region head, consistent with the substrate-axis reading. Our

Table 4. External-baseline comparison at the headline scale (FL, CA, TX). Category block reports macro-F1; region block reports Acc@10. Italicised rows are matched-head single-task ceilings on the Check2HGI substrate; the proposed row is the cross-attention multi-task model. POI-RGNN and MHA+PE are our user-disjoint reproductions; the original POI-RGNN evaluation [1] used non-user-disjoint folds, so absolute values are not directly comparable. ReHDM and the faithful STAN reproduction are reported at FL only above the headline scale (the ReHDM dual-level hypergraph scales quadratically in region cardinality, exceeding our compute budget at CA/TX).

	Category macro-F1			Region Acc@10		
Method	FL	CA	TX	FL	CA	TX
External baselines (faithful reproductions)						
Markov-1 (floor)	23.98±1.22	19.97±1.77	18.17±1.38	65.05±0.93	52.09±0.80	54.94±0.46
POI-RGNN [1]	34.49±1.14	31.78±0.82	33.03±0.70	–	–	–
MHA+PE [21]	32.06±0.23	29.13±0.71	29.91±0.43	–	–	–
STAN [9] (HGI)	–	–	–	**73.58±0.43**	**60.45±0.97**	**62.70±0.37**
ReHDM [6]	–	–	–	65.68±0.26	–†	–†
Matched-head single-task ceilings (Check2HGI substrate)						
STL category ceiling	67.16±0.13	62.39±0.13	63.11±0.13	–	–	–
STL region ceiling	–	–	–	70.62±0.09	56.85±0.09	59.44±0.09
Proposed cross-attention multi-task model						
MTL (ours)	**68.56±0.79**	**64.07±0.14**	**65.00±0.11**	63.27±0.10	47.35±0.11	42.84±0.14

Bold cells: best per column. The row label **MTL (ours)** marks the proposed model. † ReHDM not reported at CA/TX: the dual-level hypergraph scales quadratically in region cardinality, exceeding our compute budget.

multi-task row trails STAN-on-HGI by −10 to −20 pp on Acc@10, consistent with the substrate gap compounded by the cross-attention region cost.

6 Mechanism and Ablations

This section turns the reattribution into a mechanism reading, stress-tests the next-region cost against drop-in alternatives, and closes with a methodological note. Section 6.1 isolates per-visit context and replicates the decomposition at a second state. Section 6.2 reports adaptive balancers and a head-side fix that fail to close the gap. Section 6.3 states the cross-attention loss-side pitfall and the encoder-frozen protocol that decomposes it cleanly.

6.1 Per-visit context counterfactual (AL + AZ)

We isolate per-visit context through a pooled-vs-canonical counterfactual. POI-pooled *Check2HGI* mean-pools canonical vectors per POI across all check-ins and applies the per-place result uniformly to every visit. This kills per-visit variation while preserving the training signal: canonical vs. pooled isolates per-visit context, pooled vs. HGI isolates the contrastive training signal.

Under the matched-head GRU single-task ceiling at AL, canonical Check2HGI reaches 40.76 macro-F1, POI-pooled reaches 29.57, HGI reaches 25.26 (Figure 2).

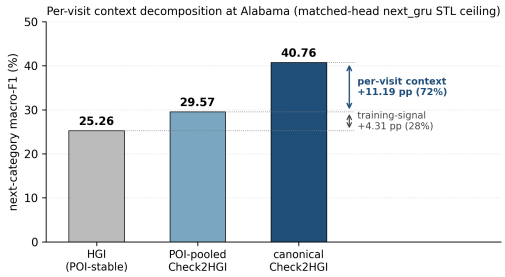

Fig. 2. Per-visit context decomposition at Alabama under the matched-head GRU single-task ceiling. The substrate gap (canonical Check2HGI − HGI = +15.50 pp) splits into a per-visit-context component (canonical − POI-pooled = +11.19 pp, ∼72 %) and a training-signal component (POI-pooled − HGI = +4.31 pp, ∼28 %). The bulk of the gap comes from per-visit variance, the property per-place embeddings cannot supply and that pooling to region cardinality smooths away.

The per-visit component is +11.19 pp (∼72 % of the substrate gap), the training-signal component +4.31 pp (∼28 %). The split survives a head-free linear probe (30.84 vs. 23.20 vs. 18.70, per-visit share ∼63 %); the GRU head amplifies the per-visit share rather than creating it.

We replicate the same three matched-head cells at AZ on identical user-disjoint folds and seed: canonical 43.17 ± 0.28, POI-pooled 34.09 ± 0.63, HGI 28.99 ± 0.51. The decomposition reproduces with both components positive: per-visit +9.08 pp (∼64 %) and training-signal +5.10 pp (∼36 %), turning the AL evidence into a two-state replicated mechanism with per-visit dominant at both states.

The exact split is not generalised to the remaining three states. The mechanism explains *why* the substrate carries the easier head and trails on the harder one: what it adds at AL/AZ is exactly what the region head does not need.

6.2 Drop-in MTL ablations do not recover the next-region cost

The next-region cost is sign-consistent across all five states (7 to 17 pp on Acc@10). To probe whether the cost is structural or recipe-borne, we ran three drop-in replacements at FL against the same folds and seeds: a magnitude balancer (FAMO [8]), a direction-aligning balancer (Aligned-MTL [14]), and a head-side fix (hierarchical-additive-softmax region head). Each is drop-in: substrate, backbone, optimiser, and schedule fixed at Section 3. None reaches a +3 pp Acc@10 recovery target at FL, and none is paired-Wilcoxon significant at $p_{\text{greater}} < 0.10$ (FAMO closes +0.62 pp; Aligned-MTL ties at −0.11 pp on next-region and regresses −0.90 pp on next-category; the hierarchical head reaches ≈ +1.7 pp dropping a single collapsed fold-init). Across the three replacements the next-region cost stays effectively intact, suggesting it is structural to the cross-attention interaction with the region head rather than an artefact of the

loss recipe. The panel ran at FL only; we cite the sign-consistency of the cost across the five states as supporting context, not as direct multi-state evidence.

6.3 Cross-attention $\lambda = 0$ is not encoder isolation

A natural ablation zeros one task's weight in Eq. (2), on the reading that $\lambda_{\text{cat}} = 0$ or $\lambda_{\text{reg}} = 0$ ablates the corresponding encoder. Under cross-attention this is unsound: the targeted encoder still updates through the key/value projections of the surviving stream, so its parameters co-adapt to the surviving task. We therefore use **encoder-frozen isolation** (targeted encoder random-init, frozen) as the clean architectural decomposition. The note generalises to any cross-task attention multi-task model that ablates by zeroing a task weight; regression tests are in our anonymous code release.

7 Discussion and Limitations

We synthesise Sections 5 and 6 into three readings: substrate task-asymmetry, limitations, and follow-up directions.

The headline reading is that the multi-task picture is dominated by the *substrate's task-asymmetric value*, not cross-task transfer through joint training. Per-visit context is paper-grade significant for next-category at every state (+14.5 to +29 pp matched-head); on next-region the same substrate ties or marginally trails per-place HGI by 1.6 to 3.1 pp under the matched-head single-task ceiling, with TOST $\delta = 2$ pp passing at California and Texas. With the substrate fixed, the cross-attention multi-task backbone adds a small category lift at four of five states (Arizona, Florida, California, Texas) and is small-significantly negative at Alabama ($\Delta = -0.78$ pp, $p = 0.036$, $n = 20$, magnitude under 2 % relative); on next-region it pays a sign-consistent 7 to 17 pp cost on Acc@10 at every state. The conventional reading where joint training transfers signal to lift both heads does not hold on the harder task: the category lift survives the substrate ablation, and the region cost survives the substrate fix. The honest reattribution is substrate first (task-asymmetric), multi-task coupling second (tradeoff-paying).

Four limitations apply. The per-visit decomposition replicates at Alabama and Arizona but does not generalise the exact split to the remaining three states. At Florida the next-region task sits in a Markov-saturated Acc@10 regime: the Markov-1-region floor reaches 65.05 Acc@10 and the multi-task backbone trails it on Acc@10 (63.27) while remaining paper-grade positive on Δ_m MRR (+2.33 %, $n = 25$); we report MRR alongside. The drop-in panel ran at Florida only and cites sign-consistency across the five states as supporting context, not as a five-state ablation. External coverage is asymmetric: ReHDM [6] is reported at Alabama, Arizona, Florida only, because its dual-level hypergraph scales quadratically with region cardinality; our POI-RGNN [1] reproduction enforces user-disjoint folds where the published evaluation used non-user-disjoint folds, so absolute gaps to the original numbers are not directly comparable.

8 Conclusion

We tested whether check-in-level contextual graph embeddings carry the multi-task POI prediction win on next-category and next-region across five U.S.-state Gowalla splits. The substrate is task-asymmetric: per-visit context lifts matched-head single-task next-category macro-F1 by +14.5 to +29 pp at every state, while per-place embeddings tie or marginally exceed it on next-region; a pooled-vs-canonical counterfactual replicates at Alabama and Arizona, with per-visit context accounting for ∼72 % and ∼64 % of the category gap. With the substrate fixed, joint multi-task training adds a small category lift at four of five states and pays a sign-consistent 7 to 17 pp region cost, the textbook tradeoff; drop-in balancers and a head-side fix do not close the gap. The multi-task picture is dominated by the substrate's task-asymmetric value, not by cross-task transfer.

Three open lines follow: asymmetric routing through PLE [16] or Cross-Stitch Networks [11] to relax shared-backbone interference; encoder enrichment fusing per-visit contrast with the explicit spatial-temporal sub-encoders of [13]; and a region-side dynamic prior conditioned on the user's recent trajectory. Recovering next-region performance without sacrificing the next-category lift is the open challenge this paper leaves to follow-up work.

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